Grace Bezold

Washington, DC
Google Scholar | LinkedIn | GitHub

Education

Johns Hopkins University — Baltimore, MD
M.S. in Artificial Intelligence (part-time) — *2024 – Present*

University of Notre Dame — Notre Dame, IN
B.A. in Computer Science and Philosophy, cum laude — *2020 – 2024*

- GPA: 3.92
- **iTREDS Scholar**, Lucy Family Institute for Data & Society

  NSF-funded two-year interdisciplinary research cohort in ethical data science, completed team capstone on analyzing effects of childhood lead exposure in South Bend, IN

Publications and Presentations

Conference Papers

1. H. Wu, S. Tian, A. Bhatta, J. Gutierrez, **G. Bezold**, G. Argueta, K. Ricanek, M. C. King, K. W. Bowyer. *Goldilocks Test Sets for Face Verification*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026, pp. 35504–35513.
2. G. Pangelinan, **G. Bezold**, H. Wu, M. C. King, K. W. Bowyer. *Lights, Camera, Matching: The Role of Image Illumination in Fair Face Recognition*. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025.
3. H. Wu, **G. Bezold**, A. Bhatta, K. W. Bowyer. *Logical Consistency and Greater Descriptive Power for Facial Hair Attribute Learning*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.
4. H. Wu, **G. Bezold**, M. Günther, T. Boult, M. C. King, K. W. Bowyer. *Consistency and Accuracy of CelebA Attribute Values*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, 2023.

Book Chapters

1. G. Pangelinan, K. S. Krishnapriya, V. Albiero, **G. Bezold**, K. Zhang, K. Vangara, M. C. King, K. W. Bowyer. *Exploring Causes of Demographic Variation in Face Recognition Accuracy*. In M. A. R. Ahad, U. Mahbub, M. Turk, & R. Hartley (Eds.), *Computer Vision: Challenges, Trends, and Opportunities* (pp. 61–81). CRC Press.

Conference Presentations

1. **G. Bezold**, R. Hingorani, B. S. Robinson, H. Yi, K. Kabotyankski, S. Rajesh, T. Merk, A. Ghazavi, A. Alijanpourotaghsara, S. Chilukuri, G. Witten, K. Mirpour, T. Fraczek, L. Jivnani, B. A. Wester, M. S. Fifer, W. K. Goodman, N. Pouratian, S. A. Sheth, N. R. Provenza. *Behaviorally Informed Neural Subspaces from Natural Conversation Predict Depressive Symptom States*. Society for Neuroscience (SfN) Annual Meeting, 2026. (presenting author)

In Preparation

1. T. Merk, **G. Bezold**, R. Hingorani, A. D. Wiese, B. Robinson, H. G. Yi, J. Altman, M. Ochoa, T. Hamre, V. Belavadi, M. Avendano-Ortega, S. Soubra, G. Reyes, T. Fraczek, E. A. Storch, W. K. Goodman, S. A. Sheth, & N. R. Provenza. *Distinct Neural Representations Encode Psychiatric State Across Multiple Timescales*.

Research Experience

Johns Hopkins University Applied Physics Laboratory — Laurel, MD
Applied ML Research Scientist, Neuroscience Group — *2024 – Present*

Behavioral Quantification and Neural State Estimation

PI: Dr. Nicole Provenza, Baylor College of Medicine, Dept. of Neurosurgery
Provenza Lab Research Goal: Investigate relevant brain-behavior relationships underlying psychiatric symptoms to develop personalized neuromodulation strategies that will improve outcomes for psychiatric disorders

- Built an end-to-end Facial Action Unit pipeline quantifying facial behavior from naturalistic conversations, and used dimensionality reduction and predictive modeling to relate expression dynamics to symptom trajectories in patients with treatment-resistant psychiatric disorders
- Derived behaviorally informed neural subspaces from intracranial recordings and showed they predict depressive symptom state
- Analyzed multimodal LLMs on video where speech and facial expression were semantically incongruent, finding that these models weight speech transcript over visual evidence and audio features
- Built a visual analytics dashboard profiling algorithm performance across video quality regimes, giving PIs a quantitative basis for assessing data usability

Egocentric Video Understanding for Mass-Casualty Triage

- Built an injury and treatment classification pipeline for egocentric medic video using frozen DINOv3 representations, for automatic documentation of casualty care
- Designed evaluation metrics for real-time injury and treatment recognition from egocentric video, assessing whether model outputs are timely and reliable enough to support a medic during mass-casualty triage

Eye-Tracking and Cognitive Load

- Developed a MATLAB package for cognitive-load estimation from eye tracking equipment, included feature extraction over fixations, saccades, and pupil traces, with time-series modeling
- Designed test protocols, collected data from submarine operators, and ran the resulting cognitive-load analysis

Computer Vision Research Lab (CVRL), University of Notre Dame — Notre Dame, IN
Undergraduate Research Assistant, Advisor: Prof. Kevin W. Bowyer — *2021 – 2024*

Performed experiments isolating the causes of demographic variation in face recognition accuracy, and benchmark work asking whether standard evaluations measure what they are assumed to measure.

- Senior research project showing that balancing facial brightness distributions reduces the face matching accuracy gap between demographic groups, identifying an image acquisition factor, instead of the face recognition model itself, as a contributor to observed disparities. Collaborated on the work from initial question through experimental design, analysis, figures, and manuscript (WACV 2025)
- Contributed to benchmark and label-validity work establishing that widely used attribute annotations are internally inconsistent, that facial hair attributes admit a logically consistent taxonomy, and that verification test sets can be made harder through targeted pair construction rather than image degradation (CVPR 2023 ×2, CVPR 2026)

Industry Experience

Mars, Incorporated — Chicago, IL
Next Generation Technology Intern — *Summer 2023*

- Extended a deep learning pipeline interpreting dental radiographs for diagnostic support in Mars veterinary hospitals
- Trained Stable Diffusion models to synthesize thousands of segmentation masks for augmenting a limited dental imaging dataset

Mentoring

- **Research Mentor**, JHU Applied Physics Laboratory, Summer 2026: Mentored a summer intern on our project of using egocentric video for automatic documentation of triage. I discussed goals with the student, scoped a project and research question in the domain they wanted to explore, provided background learning information, developed the research question, advised on analysis and reviewed code.
- **Volunteer Tutor**, Serve Your City, Washington, DC, 2025 – Present
- **Calculus Tutor**, University of Notre Dame, 2021 – 2023

Service and Outreach

- **Philosophy Section Editor**, *Undergraduate Journal of the College of Arts and Letters*, University of Notre Dame, 2023 – 2024
- **Intern**, Accompanying Returning Citizens with Hope (ARCH), Columbus, OH, Summer 2021 — advocated for bipartisan bail reform legislation and testified before the Ohio State Senate

Technical Skills

- **Languages:** Python, MATLAB, C
- **ML:** PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, pandas, CEBRA, Hugging Face
- **Infrastructure:** Git, Linux, Docker, SLURM and use of HPC clusters

Selected Coursework

- **Mathematics & CS Theory:** Linear Algebra and Introduction to Differential Equations, Probability and Statistics, Discrete Mathematics, Design and Analysis of Algorithms, Theory of Computing
- **Machine Learning:** Machine Learning and Neural Networks, Applied Machine Learning, Biometrics, Creating AI-Enabled Systems, AI in Healthcare
- **Philosophy:** Cognitive and Behavioral Foundations for AI, Philosophy of AI, Advanced Formal Logic, Formal Logic